

Data Processing Procedure (DPP)

Objective: Hydrogen Deflagration Time-Series Analysis.

Procedure Steps

Data Acquisition and Import

Collect Raw Data

Record the pressure data from the deflagration test at a known sampling rate and store all relevant metadata (e.g., test conditions, sensor specifications).

Import into MATLAB

Load the raw time-series data (e.g., using load or CSV import) and verify the correct time alignment, sampling frequency, and units.

Outcome

A raw pressure signal and associated metadata are available in MATLAB.

Matlab Startup File

This m-File code prepare the enviroment to provide access to auxilliary files and sub-folders.

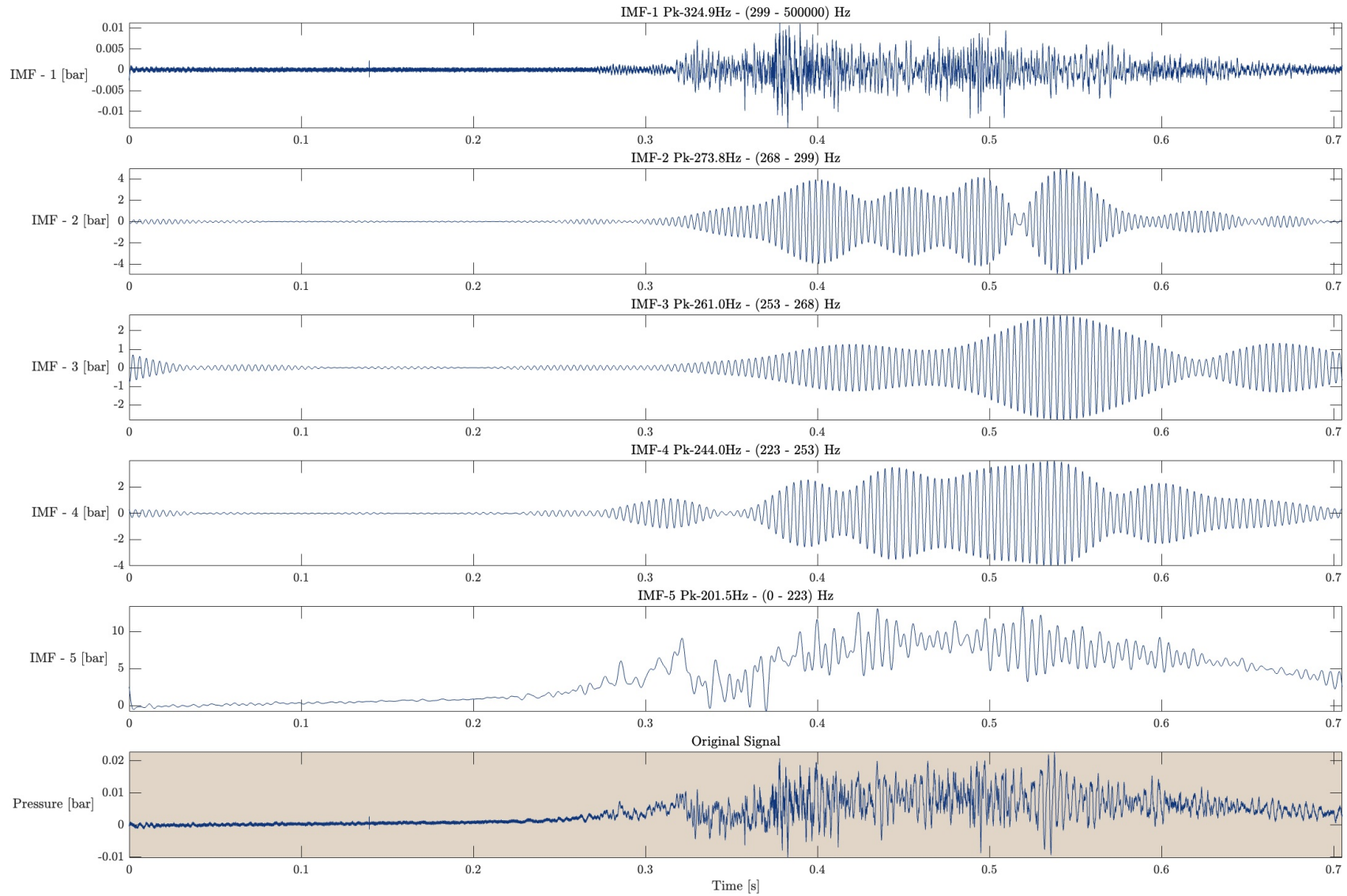
Import Signal Data

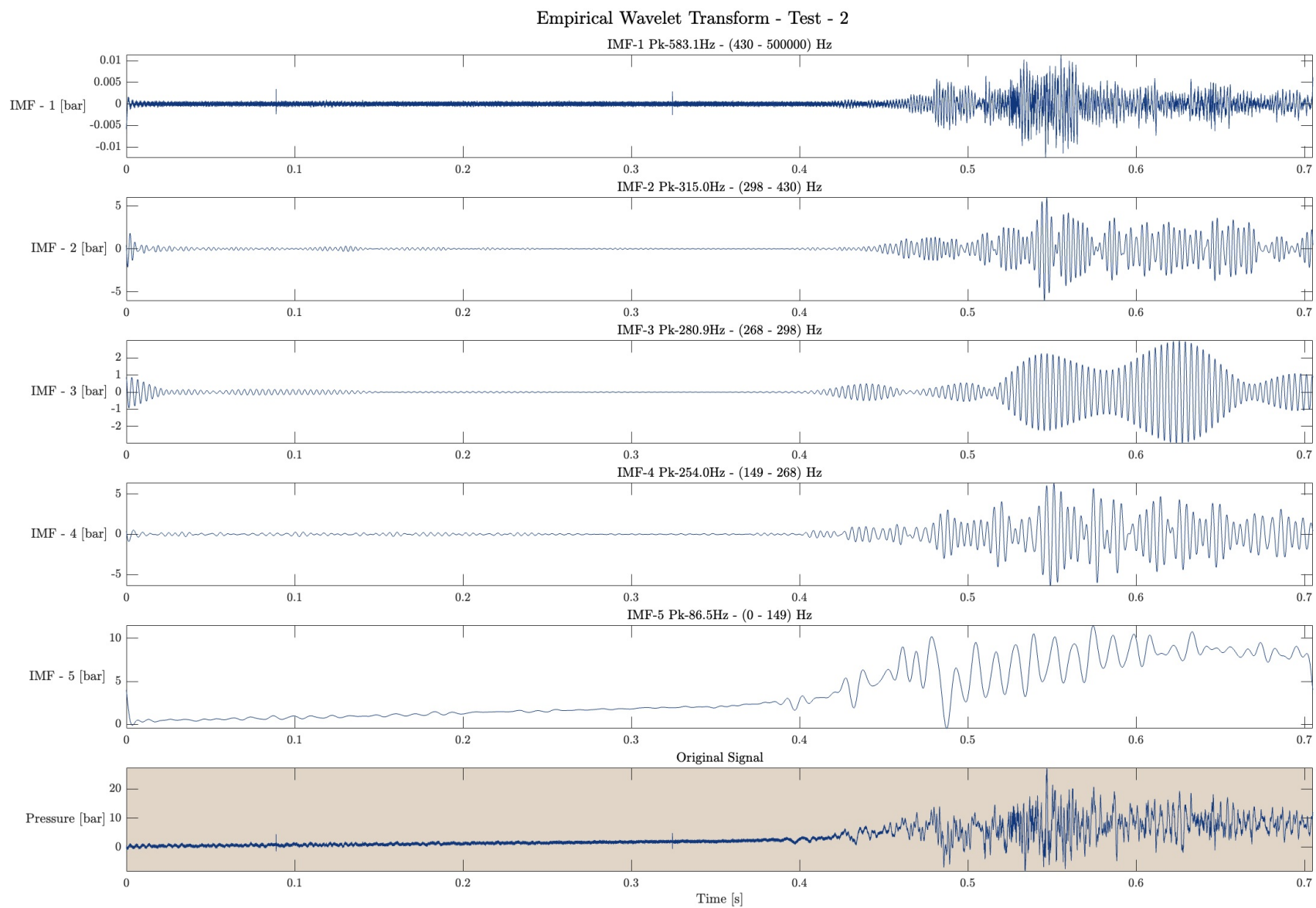
Load "Raw" Signal

Note: Details on DAQ system setting are required, to check if any hardware filters has been used.

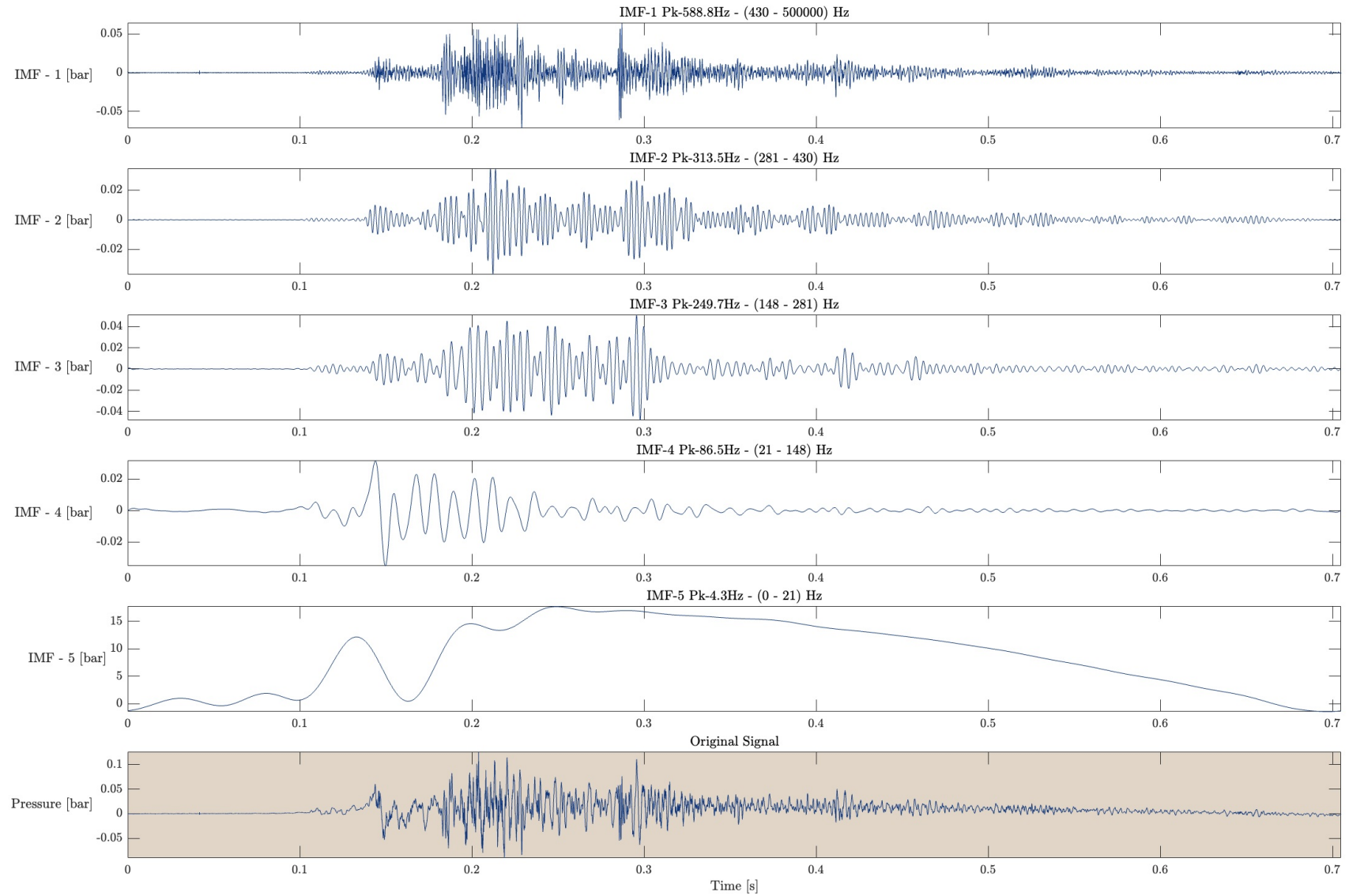
Use ewt to obtain a multiresolution analysis (MRA) of the signal, and the corresponding analysis coefficients.

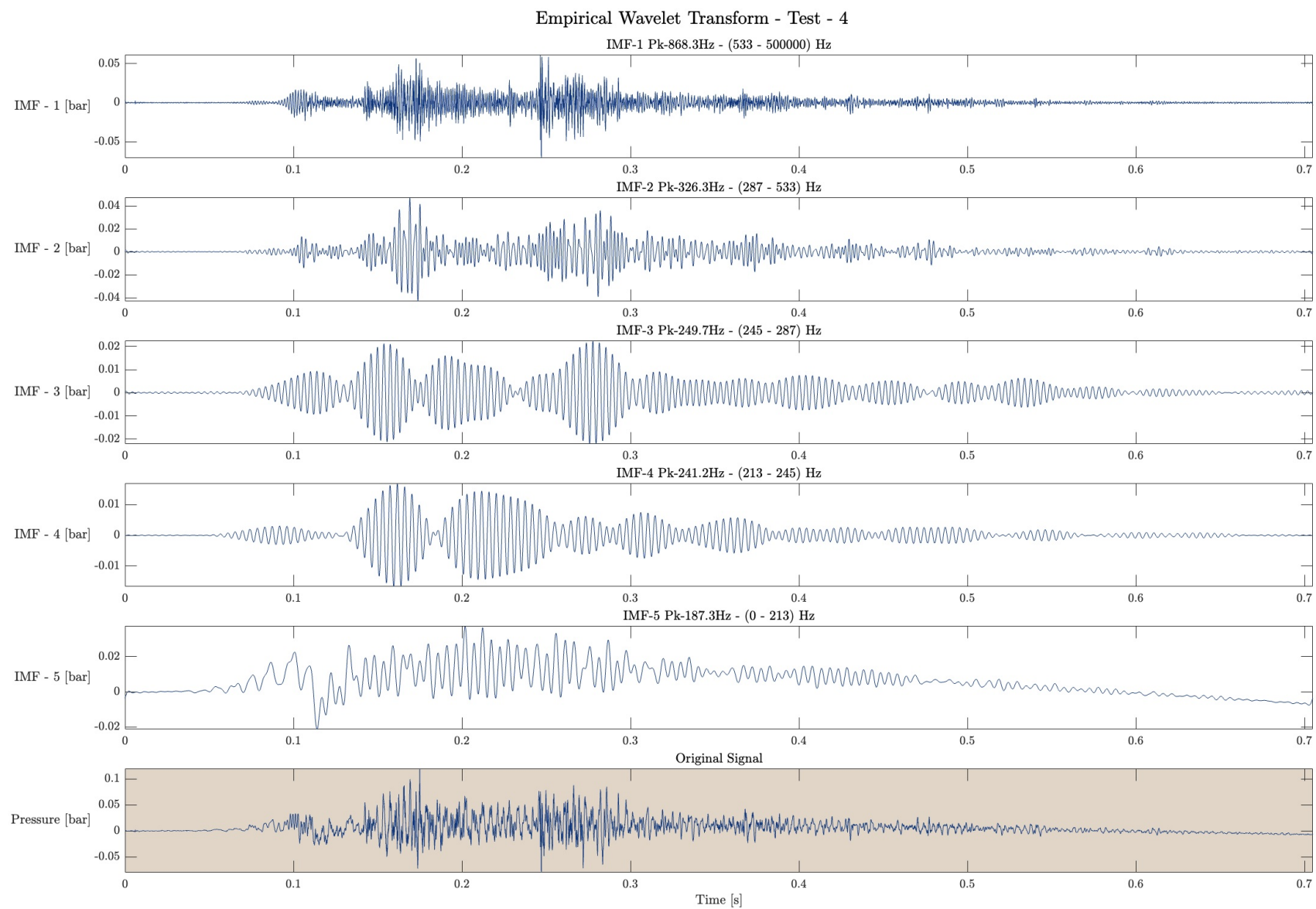
Empirical Wavelet Transform - Test - 1



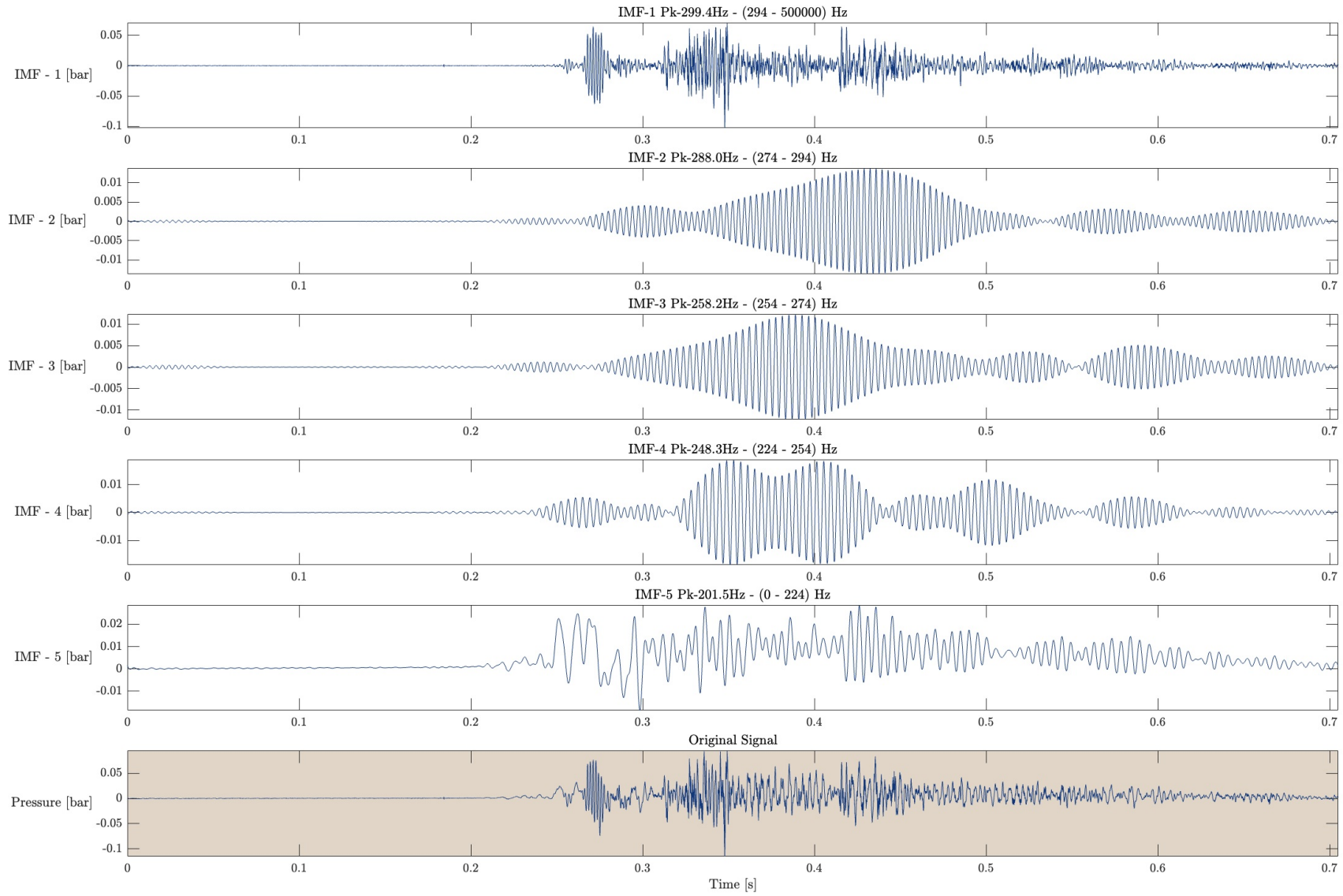


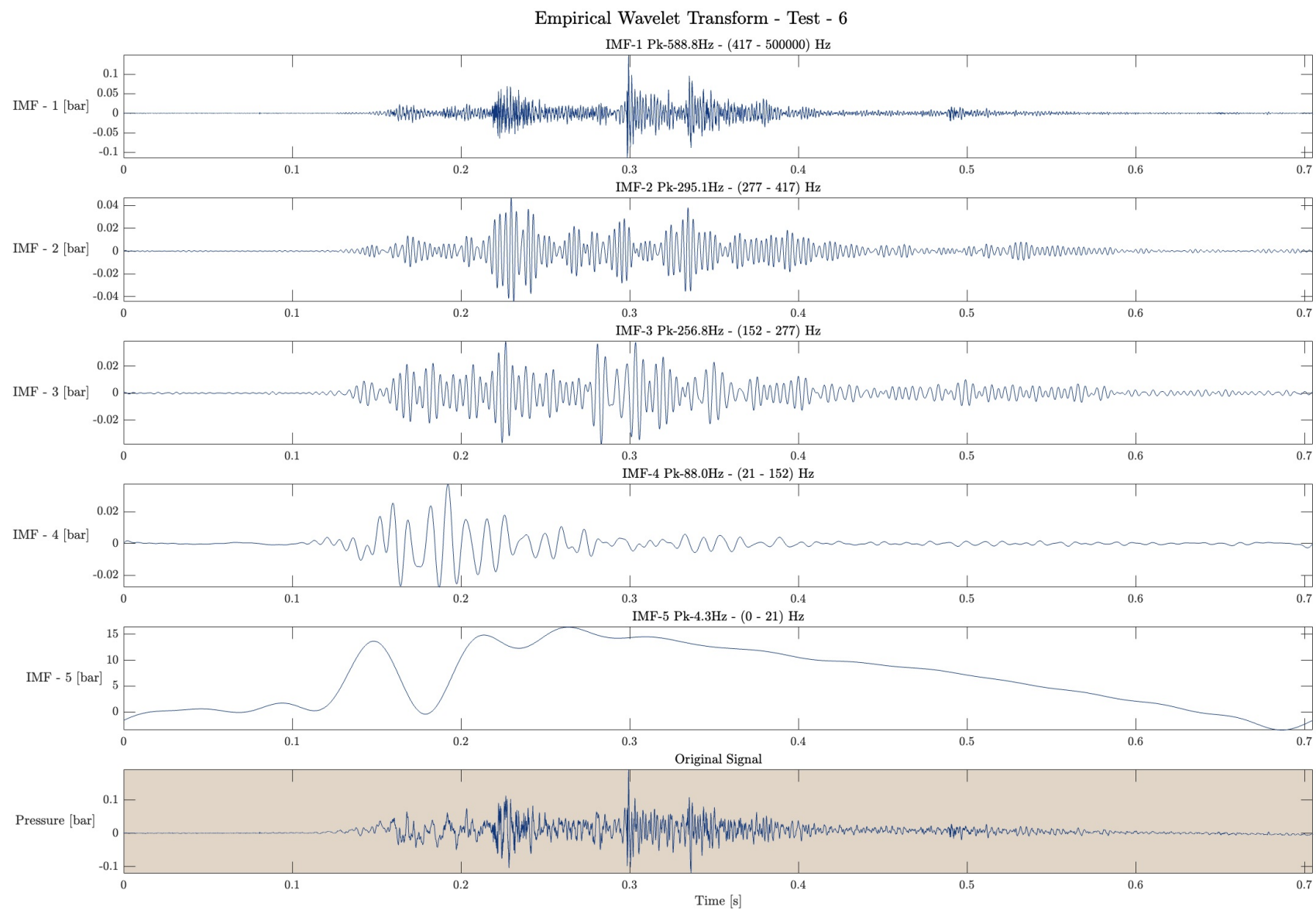
Empirical Wavelet Transform - Test - 3

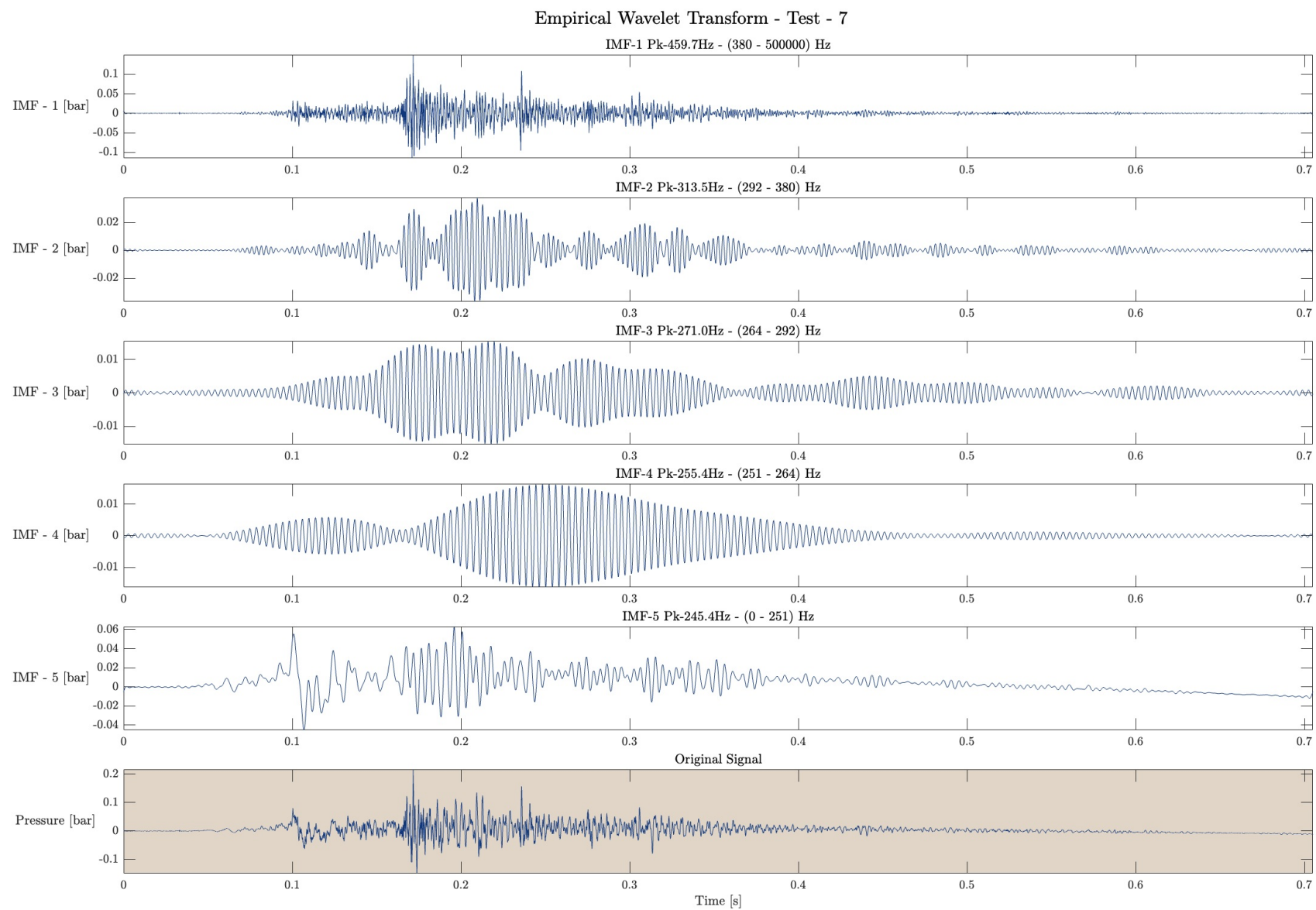




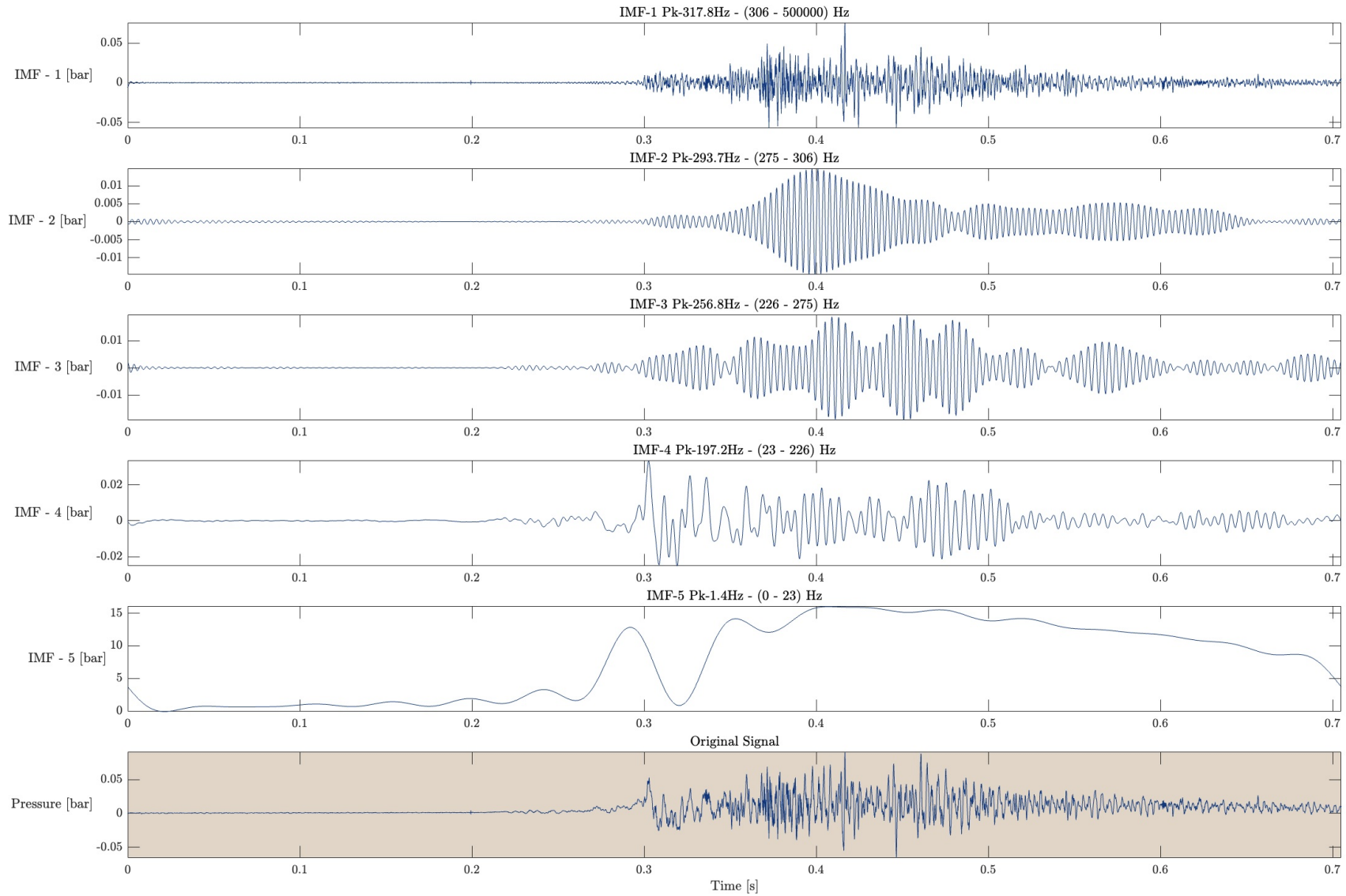
Empirical Wavelet Transform - Test - 5



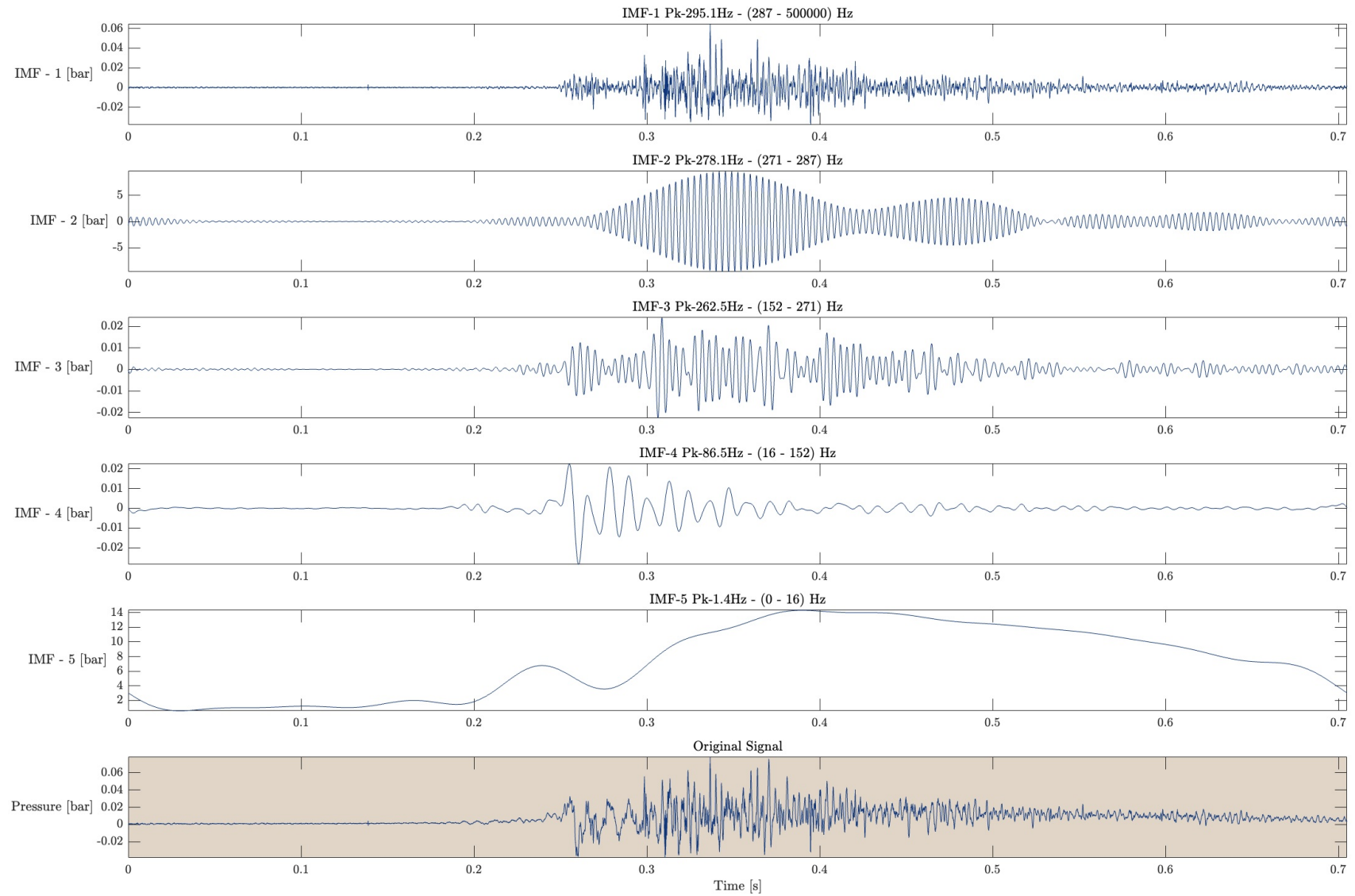




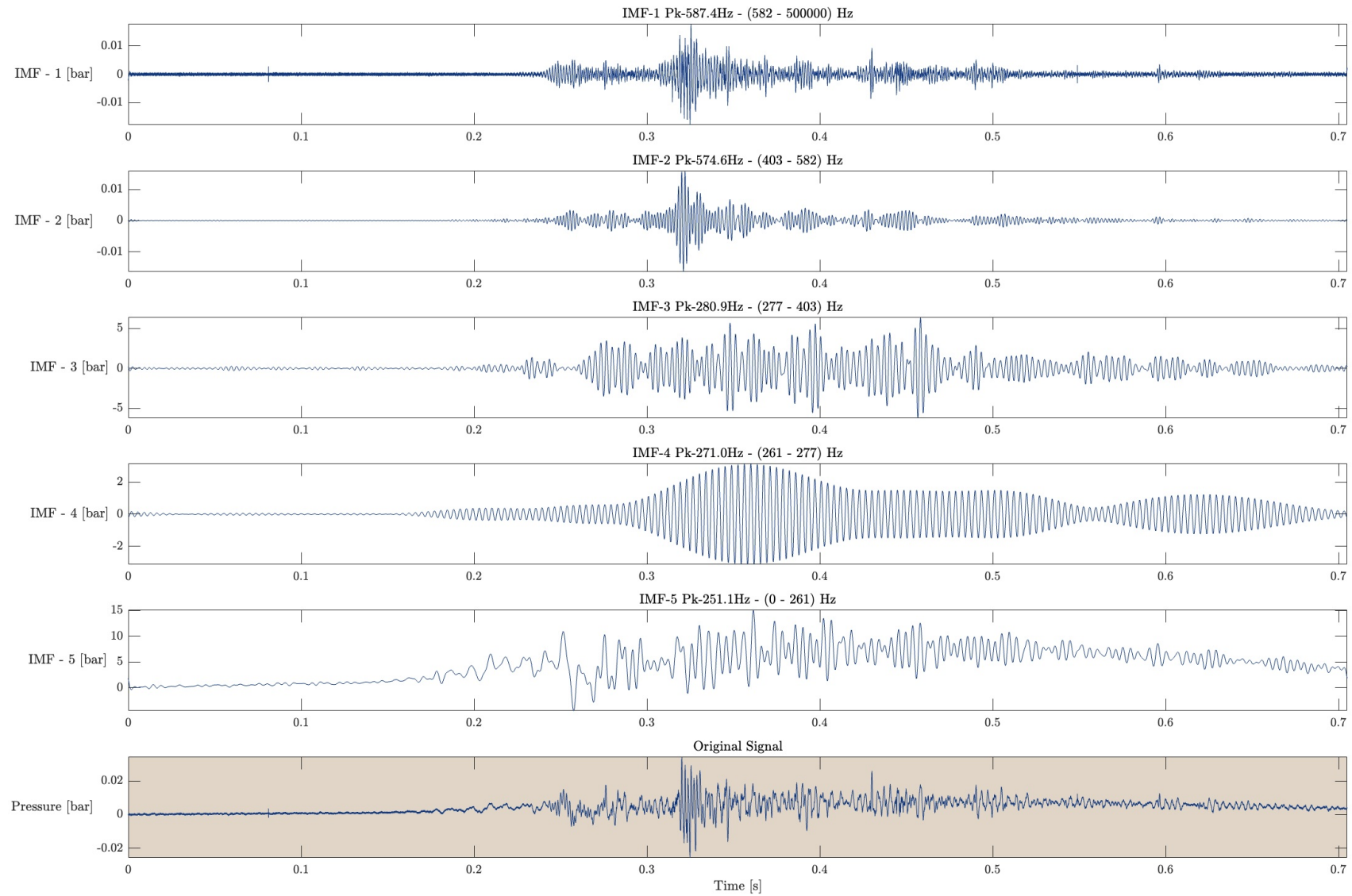
Empirical Wavelet Transform - Test - 8



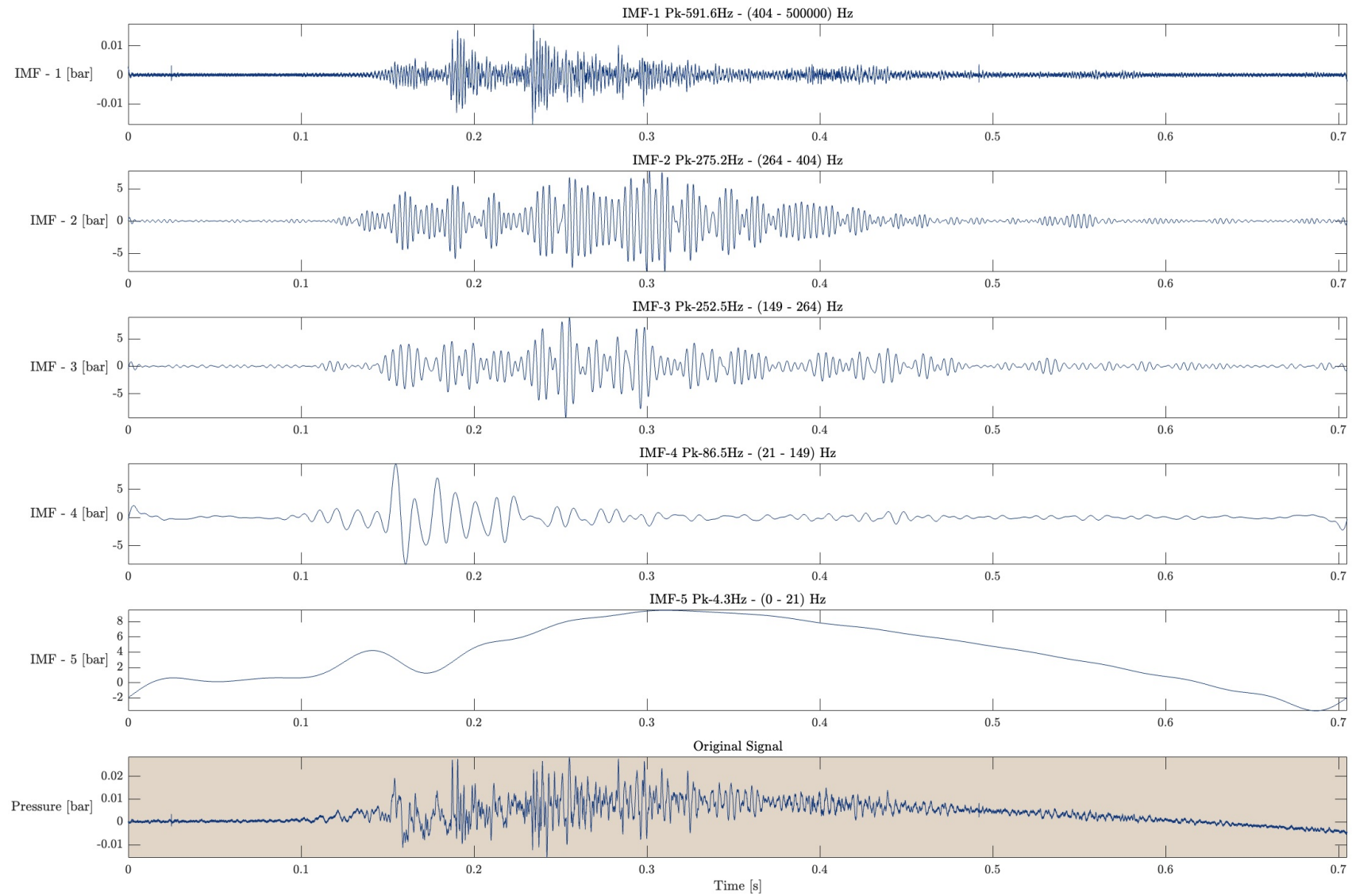
Empirical Wavelet Transform - Test - 9



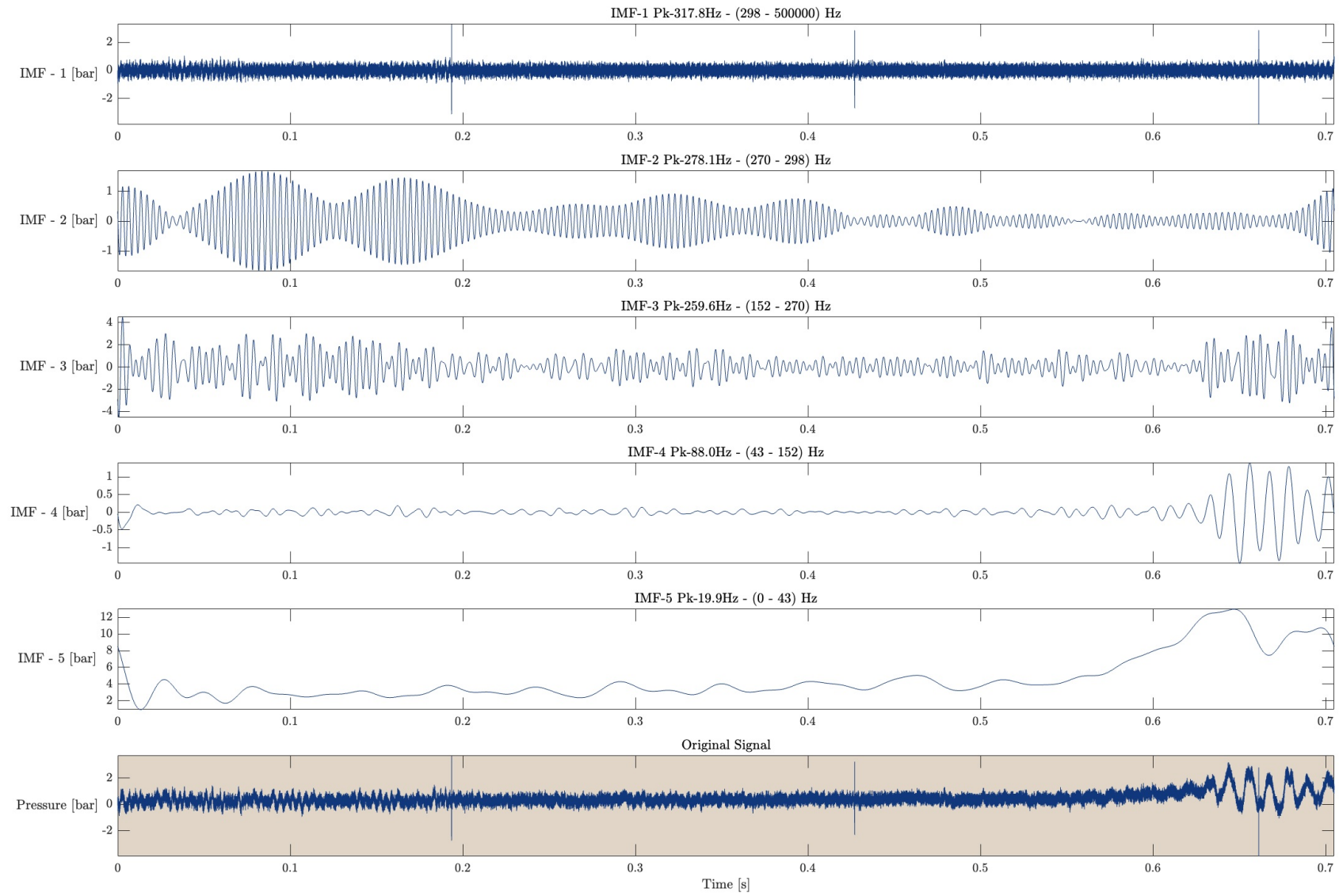
Empirical Wavelet Transform - Test - 10



Empirical Wavelet Transform - Test - 11



Empirical Wavelet Transform - Test - 12



Test Number	Mode Number	Peak Frequency (Hz)	Passband Start (Hz)	Passband End (Hz)
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1	1	324.89	299.35	5e+05
1	2	273.82	268.14	299.35
1	3	261.05	252.53	268.14
1	4	244.02	222.74	252.53
1	5	201.46	0	222.74
2	1	583.1	429.88	5e+05
2	2	314.96	297.93	429.88
2	3	280.91	268.14	297.93
2	4	253.95	148.97	268.14
2	5	86.543	0	148.97
3	1	588.78	429.88	5e+05
3	2	313.54	280.91	429.88
3	3	249.7	147.55	280.91
3	4	86.543	21.281	147.55
3	5	4.2562	0	21.281
4	1	868.27	533.44	5e+05
4	2	326.31	286.58	533.44
4	3	249.7	245.44	286.58
4	4	241.19	212.81	245.44
4	5	187.27	0	212.81
5	1	299.35	293.68	5e+05
5	2	288	273.82	293.68
5	3	258.21	253.95	273.82
5	4	248.28	224.16	253.95
5	5	201.46	0	224.16
6	1	588.78	417.11	5e+05
6	2	295.1	276.65	417.11
6	3	256.79	151.8	276.65
6	4	87.962	21.281	151.8
6	5	4.2562	0	21.281
7	1	459.67	380.22	5e+05
7	2	313.54	292.26	380.22
7	3	270.98	263.88	292.26
7	4	255.37	251.12	263.88
7	5	245.44	0	251.12
8	1	317.8	306.45	5e+05
8	2	293.68	275.23	306.45
8	3	256.79	225.58	275.23
8	4	197.2	22.7	225.58

8	5	1.4187	0	22.7
9	1	295.1	286.58	5e+05
9	2	278.07	270.98	286.58
9	3	262.47	151.8	270.98
9	4	86.543	15.606	151.8
9	5	1.4187	0	15.606
10	1	587.36	581.68	5e+05
10	2	574.59	402.92	581.68
10	3	280.91	276.65	402.92
10	4	270.98	261.05	276.65
10	5	251.12	0	261.05
11	1	591.61	404.34	5e+05
11	2	275.23	263.88	404.34
11	3	252.53	148.97	263.88
11	4	86.543	21.281	148.97
11	5	4.2562	0	21.281
12	1	317.8	297.93	5e+05
12	2	278.07	269.56	297.93
12	3	259.63	151.8	269.56
12	4	87.962	42.562	151.8
12	5	19.862	0	42.562